
Research on Clue-driven Green Food Purchase Intention Mechanism: Empirical Evidence from China Consumers

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Abstract: With the rise of green consumption concepts, green food has gradually become an important area of consumer concern. However, existing research still lacks systematic exploration of the mechanisms by which consumers rely on external cues to form green food purchase intentions. Based on cue utilization theory and the S-O-R model, this paper constructs a theoretical framework for how external cues influence green food purchase intentions, introduces perceived value as a mediating variable and consumer attitudes as a moderating variable to reveal the cue-driven green consumption decision-making mechanism. Through questionnaire surveys conducted among online consumers in China, 305 valid samples were obtained, and structural equation modeling was employed for empirical analysis. The results show that different types of external cues significantly positively influence green food purchase intentions; perceived value partially mediates the relationship between external cues and purchase intentions; meanwhile, consumer attitudes positively moderate the impact of external cues on perceived value. This study deepens the understanding of green consumption behavior formation pathways from the perspective of cue mechanisms and provides theoretical foundations and practical insights for green food enterprises to formulate marketing strategies.

Keywords: Cue Theory; Green Consumption; Perceived Value; Consumer Attitude; Purchase Intention

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1 Introduction

With the continuous deepening of sustainable development concepts, green consumption has gradually become an important topic in consumer behavior and marketing research. As an essential component of green consumption, green food has attracted increasing attention from consumers due to its health and environmental attributes. In recent years, China's green food industry has experienced rapid development, with the market size continuously expanding and consumers' concern over food safety and environmental protection issues growing steadily. Against this backdrop, exploring the formation mechanism of Chinese consumers' green food purchase intentions has become a crucial research direction for understanding green consumption behavior.

Existing research primarily explains green consumption behavior through perspectives such as environmental awareness, health motivations, and social norms. However, the mechanisms by which consumers rely on external information cues to form purchasing decisions under incomplete information scenarios remain underexplored. According to cue utilization theory, when product quality cannot be directly observed, consumers often depend on available information signals to assess product value. This mechanism is particularly crucial in the context of green food, as the production processes and quality characteristics of green food are highly concealed. Consequently, external cues such as brand names, certification labels, price information, and product reviews play a pivotal role in consumer decision-making.

However, existing research predominantly focuses on the impact of single cues on green consumption behavior, lacking systematic analysis of integrated mechanisms involving multiple cues. Additionally, the psychological processing pathways consumers employ when interpreting external cues remain inadequately explained, particularly regarding the mechanisms through which perceived value and consumer attitudes function as psychological variables. Building on this gap, this study constructs a theoretical model of external cues influencing green food purchase intention within the Stimulus-Organism-Response (S-O-R) framework. We focus on examining the mediating role of perceived value between external cues and purchase intention, while further analyzing the moderating effect of consumer attitudes in the relationship between external cues and perceived value. These findings aim to elucidate the cue-driven mechanisms underlying green consumption decision-making processes.

This paper takes China consumers as the research object, collects data through questionnaire surveys, and conducts empirical analysis using structural equation modeling. The research findings not only contribute to deepening the theoretical understanding of the formation mechanism of green consumption behavior, but also provide practical insights for green food enterprises to optimize their market communication and signal management strategies.

2 Theoretical Background

2.1. Literature Review

With the continuous advancement of green consumption research, academic circles have gradually shifted from macro-environmental and individual characteristics perspectives to systematic exploration of the formation mechanisms of green food consumption behaviors. Existing studies indicate that consumer preferences for green foods are significantly influenced by factors such as food safety awareness, health motivations, and product attributes. Fujita (2006) noted that with socio-economic development, increasing demand for safe and harmless food products has made green food brand building a crucial direction for agricultural development. In different national contexts, scholars have also conducted research from consumer attitude and cognitive perspectives. For instance, Bo Won Suh (2012) found that consumer attitudes, trust, and prior consumption experiences all impact green food purchasing decisions; Parichard et al. (2012) emphasized that consumers' perceptions of green food quality and pricing play a significant role in decision-making processes.

Further research reveals that factors influencing green food consumption exhibit multidimensional characteristics. Ozguven (2012) demonstrated through questionnaire data that food safety, pricing, product quality, and health considerations significantly impact consumer purchasing intentions. Khan M. et al. (2015) similarly emphasized the pivotal role of food safety

and health motivations in green food consumption. However, Ritter (2015) argued that price factors do not universally influence green consumption, with their impact being moderated by contextual variables such as income levels. These findings indicate that green consumption decisions are not driven by a single factor but result from the combined effects of multiple variables.

From the perspective of consumer psychology and value orientation, relevant studies further reveal the complexity of green consumption behaviors. Teng (2016) found that health awareness, food safety concerns, and eco-friendliness are significantly positively correlated with green food purchasing behavior. Chemat et al. (2017) suggested that increased purchasing power would further drive the growth of the green food market. Torben (2018) discovered that consumers' alignment with health and environmental values significantly promotes green consumption.

In the context of China, scholars have primarily conducted research from dimensions such as demographic characteristics, values, and consumption cognition. Wang Jianguo (2004) analyzed green food consumption preferences from a values perspective; Li Jianxin (2006) pointed out that demographic variables have relatively limited explanatory power for green consumption behavior; Yang Zhi and Dong Xuebing (2010) argued that income and education level significantly influence green product selection. Additionally, Tang Xueyu et al. (2010) found through empirical analysis that consumption motivation has a significant positive impact on purchasing behavior. Recent studies further indicate that factors such as consumer awareness, product certification, price, and quality all affect green food consumption behavior.

Overall, existing research has examined factors influencing green food purchase intention from multiple perspectives including individual characteristics, psychological motivations, and product attributes, while noting potential discrepancies between consumer attitudes and actual behaviors. Factors such as product pricing, brand image, and information perception demonstrate differentiated impacts across different contexts. However, current studies still exhibit two limitations: first, most research focuses on single influencing factors while insufficient attention is paid to the synergistic mechanisms of multiple information cues; second, explanations regarding how consumers form purchase intentions through psychological processing under information incompleteness remain inadequate. Therefore, it is essential to systematically explore the formation pathways of green food consumption decisions from the perspective of cue utilization.

2.2 Theoretical Basis

2.2.1 Thread Utilization Theory

The cue theory was first proposed by Cox (1962), which posits that consumers rely on observable informational cues when evaluating product quality [22]. Olson et al. (1972) subsequently categorized cues into internal and external types based on product attributes, with internal cues primarily related to physical characteristics and external cues encompassing brand, price, and origin information [23]. Building on this framework, Haeckel et al. (2003) further classified cues into functional cues, mechanism cues, and human-induced cues, emphasizing how multiple cues collectively shape consumer experience. Berry (2006) conducted a systematic analysis of cue roles in consumer evaluation processes from a service context perspective.

In green consumption research, scholars have examined consumer behavior mechanisms through the lens of cues. Qiu Li (2012) demonstrated that green certification and pricing factors significantly influence green brand purchase behavior [26], while Wang Xiaoting (2017) analyzed how brand perception and pricing affect online travel decision-making based on cue theory [27]. These findings collectively indicate that external cues such as green certification, brand reputation, price information, and product reviews play crucial roles in shaping consumers' green consumption decisions. Building upon existing research, this study systematically investigates the impact mechanisms of brand names, certification labels, pricing, and product quality reviews on green food purchase intention by focusing on these key external cues.

2.2.2 Perceived Value

Kolter (1972) posited that perceived value encompasses not only monetary costs but also multidimensional expenditures including time, emotional investment, and energy expenditure [28]. Subsequent research expanded this concept to incorporate functional, affective, and social dimensions. In green consumption contexts, Koller et al. (2011) introduced the notion of green

value, emphasizing the environmental benefits and social recognition gained by consumers when purchasing eco-friendly products. Accordingly, this study defines perceived value as consumers' comprehensive evaluation of green food's functional attributes, emotional experiences, and environmental benefits.

2.2.3 Consumer Attitude

Consumer attitudes are generally defined as stable psychological tendencies formed by individuals toward specific products or brands. Both rational behavior theory and planned behavior theory indicate that positive attitudes enhance the likelihood of behavioral outcomes. Existing research demonstrates that consumer attitudes play a significant predictive role in brand preference, purchase intention, and willingness to pay premium prices. In green consumption contexts, consumer attitudes may influence how individuals interpret product information cues, thereby moderating purchasing decisions.

2.2.4 Stimulus-Organism-Response (S-O-R) Theory

The Stimulus-Organism-Response model originates from the stimulus-response theory in psychology, positing that individual behavior arises as a reaction following external stimuli processed internally. In consumer behavior research, this model is widely used to explain how marketing stimuli influence purchase intent through emotional and cognitive mechanisms. For instance, Liu Y et al. applied the S-O-R model to analyze the impact of interpersonal attraction factors on purchase intent in social platforms [34]; Fu S et al. investigated the mechanism of situational similarity on online consumption behavior [35].

In green food consumption contexts, external cues such as brand names, certification labels, pricing, and product reviews can be regarded as stimuli. These stimuli influence consumers' psychological perceptions (e.g., perceived value) and subsequently affect their purchasing behaviors. Therefore, the S-O-R model provides a crucial theoretical foundation for establishing the formation mechanism of green food purchase intention in this study.

3 Research Models and Hypotheses

Building upon previous literature reviews and theoretical analyses, this study develops a research model for green food purchase intention formation mechanisms by applying cue utilization theory and integrating characteristics of green food consumption contexts. Furthermore, utilizing the Stimulus-Organism-Reaction (S-O-R) theoretical framework, we systematically examine how external informational cues influence consumers' psychological perceptions and behavioral responses.

3.1 Research Model Construction

According to cue utilization theory, consumers typically rely on observable product cues to make quality and value judgments when faced with incomplete information. In the context of green food products, where production processes and quality attributes are highly concealed, external cues play a more critical role in consumer decision-making. Therefore, this study identifies brand names, certification labels, pricing, and quality reviews as external stimulus factors (S) to characterize information signal sources in green food consumption decisions.

According to the S-O-R theory, external stimuli influence individuals' internal psychological states, thereby triggering behavioral responses. In the process of green food consumption decision-making, consumers' subjective evaluation of product value serves as a key psychological mechanism shaping purchasing intent. Therefore, this study employs perceived value as the core organism-level variable (O) to explain how external cues affect consumers' psychological perceptions.

Furthermore, under the combined influence of multiple external cues, consumers' perceived value of green foods significantly impacts their purchase intention, ultimately manifesting as purchasing behavior propensity (R). Existing research indicates that consumer attitudes, as stable psychological tendencies, affect individuals' interpretation of information cues and value assessment processes. Therefore, this study introduces consumer attitudes as a moderating variable to examine their regulatory role in the relationship between external cues and perceived value.

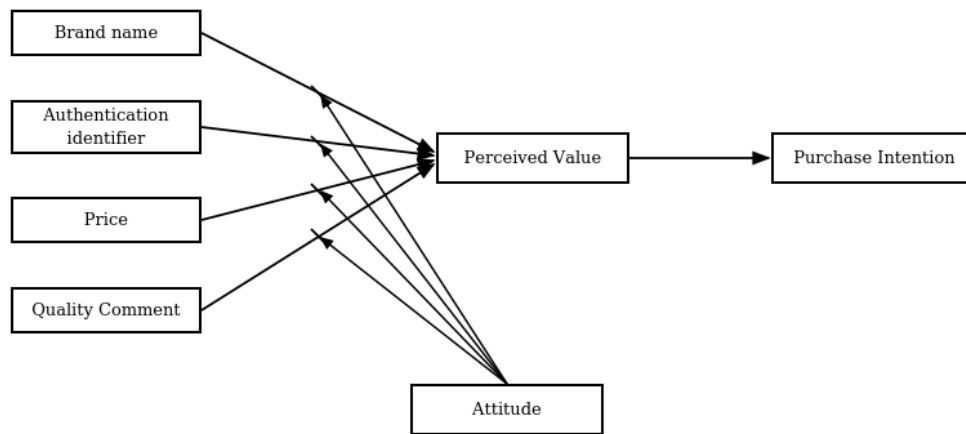


Figure 1. Research Model

3.2 Theoretical Hypotheses

External cues and purchase intention.

In external cue studies, brand names are commonly regarded as critical signals for conveying product quality and reputation. Existing research indicates that consumers often infer product quality through brand cues when product information is insufficient, thereby influencing their purchasing decisions. Therefore, this paper proposes the following hypothesis:

H1: Brand name has a positive impact on consumers' purchase intention for green food.

Certification labels serve as institutionalized signals for product quality and safety, helping to reduce consumers' cognitive uncertainty in situations of information asymmetry. Relevant studies have shown that green certification can enhance consumer trust and purchase intention. Accordingly, it is proposed that:

H2: The certification label positively influences consumers' purchase intention for green food.

Price serves as both a critical signal of product value and a key basis for consumers to infer product quality. In the context of green food consumption, consumers may perceive higher prices as a symbol of product quality and safety, thereby influencing their purchasing decisions. Therefore, it is proposed that:

H3: Price exerts a positive influence on consumers' purchase intention for green food.

With the development of digital consumption environments, product quality reviews have become a crucial source of information for consumers. High-quality reviews help reduce uncertainty and enhance consumers' perception of product value, thereby strengthening purchase intent. Based on this, the following hypothesis is proposed:

H4: Quality reviews positively influence the purchase intention for green foods.

The Mediating Role of Perceived Value

Perceived value reflects consumers' comprehensive evaluation of a product's overall benefits and costs, serving as a key psychological mechanism influencing consumption decisions. Existing studies indicate that consumers' subjective perception of product value significantly impacts their purchase intention. Additionally, external cues such as price, brand reputation, and word-of-mouth may influence consumption behavior by affecting perceived value. Therefore, it is proposed that:

H5: Perceived value positively influences the purchase intention for green foods.

H5a: Perceived value mediates the relationship between brand name and purchase intention.

H5b: Perceived value mediates the relationship between certification labels and purchase intention.

H5c: Perceived value mediates the relationship between price and purchase intention.

H5d: Perceived value mediates the relationship between quality reviews and purchase intention.

The Moderating Role of Consumer Attitude

Attitude is recognized as a crucial psychological factor influencing behavioral intentions. Both Rational Behavior Theory

and Planned Behavior Theory indicate that positive evaluations of behaviors can enhance their likelihood of occurrence. In green consumption contexts, consumer attitudes may shape how individuals interpret external cues, which subsequently influences value perception processes. Therefore, this study proposes a hypothesis regarding the moderating role of consumer attitudes in the relationship between external cues and perceived value:

H6a: Consumer attitudes mediate the relationship between brand effect and perceived value.

H6b: Consumer attitudes mediate the relationship between certification labels and perceived value.

H6c: Consumer attitudes mediate the relationship between price and perceived value.

H6d: Consumer attitudes mediate the relationship between quality reviews and perceived value.

3.3 Summary of Research Hypotheses

Table1. Summary of Research Hypotheses

variable	research hypothesis
Brand name	H1: Brand name has a positive impact on consumers' purchase intention for green food.
Authentication identifier	H2: The certification label positively influences consumers' purchase intention for green food.
price	H3: Price exerts a positive influence on consumers' purchase intention for green food.
Quality Comment	H4: Quality reviews positively influence the purchase intention for green foods.
	H5: Perceived value positively influences the purchase intention for green foods.
	H5a: Perceived value mediates the relationship between brand name and purchase intention.
perceived value	H5b: Perceived value mediates the relationship between certification labels and purchase intention.
	H5c: Perceived value mediates the relationship between price and purchase intention.
	H5d: Perceived value mediates the relationship between quality reviews and purchase intention.
	H6a: Consumer attitudes mediate the relationship between brand effect and perceived value.
consumer attitude	H6b: Consumer attitudes mediate the relationship between certification labels and perceived value.
	H6c: Consumer attitudes mediate the relationship between price and perceived value.
	H6d: Consumer attitudes mediate the relationship between quality reviews and perceived value.

4 Research Design and Empirical Results

4.1 Questionnaire Design and Variable Measurement

This study employed questionnaire surveys to collect data. The questionnaire comprised multiple subscales designed to measure key research variables including brand names, certification labels, pricing, quality reviews, perceived value, consumer attitudes, and purchase intent. To ensure the reliability and applicability of the measurement tools, we adapted existing validated scales by incorporating context-specific revisions for green food consumption scenarios. The electronic questionnaire was distributed through an online survey platform.

This questionnaire primarily consists of two sections with a total of 29 measurement items. The first section is the core measurement component, designed to assess respondents' perceptions and evaluations of various research variables in the context of green food consumption. The second section covers demographic information, including demographic characteristics such as gender, age, education level, occupation, and monthly income.

Each latent variable was measured with three or more observed items using a 5-point Likert scale, where "1" indicates "strongly disagree," "2" indicates "disagree," "3" indicates "neutral," "4" indicates "agree," and "5" indicates "strongly agree."

In terms of variable design, this study employs the Clue Utilization Theory and S-O-R theoretical framework, identifying brand name, certification labels, price, and quality reviews as external clue variables, perceived value as a mediator variable, consumer attitudes as a moderating variable, and purchase intention as the outcome variable. All variables are measured using established scales that have been locally adapted for the green food research context.

The study comprises four items for the brand name variable, referencing Wang Jun [49] and Dodds & Monroe [50]; four

items for the certification symbol variable, drawing on Husted et al. [51] and Zhang Lisheng et al. [52]; three items for the price variable, based on Mei Xue [53] and Ailawadi et al. [54]; three items for the quality review variable, incorporating Dhanasobhon [55]'s research; four items for the perceived value variable, referencing Sweeney et al. [56], Cui Dengfeng et al. [57], and Zhang Shujun [58]; three items for the consumer attitude variable, referencing Khaola et al. [59] and Barbarossa et al. [60]; and three items for the purchase intention variable, referencing Han et al. [61] and Zeithaml [43].

Overall, the scale employed in this study has clear theoretical foundations and effectively captures the core meanings of various variables in green food consumption contexts.

Table 2. Summary of Variable Measurement Sources

Variable	Number of items	Reference
Brand name	4	Wang Jun (2011); Dodds & Monroe (1985)
Authentication identifier	4	Husted et al. (2014); Zhang Lisheng et al. (2012)
Price	3	Mei Xue (2010); Ailawadi et al. (2001)
Quality Comment	3	Dhanasobhon (2007)
Perceived value	4	Sweeney et al. (2001); Cui Dengfeng et al. (2018); Zhang Shujun (2019)
Consumer attitude	3	Khaola et al. (2014); Barbarossa et al. (2015)
Willingness to buy	3	Han et al. (2009); Zeithaml (1988)

4.2 Questionnaire Distribution and Sample Recovery

To effectively obtain research samples, this study generated questionnaire links on the Wen juanxing platform and primarily distributed them through online channels such as WeChat, QQ, and Weibo, while leveraging referrals from friends and school teachers to expand the survey coverage.

Through the aforementioned methods, a total of 347 questionnaires were collected. After screening the returned questionnaires, those with abnormal response times, obvious patterns in answering behavior, incomplete information entries, or non-standard responses were excluded, resulting in 305 valid questionnaires. The number of valid samples meets the basic requirement of "5 to 10 times the number of measurement items," which is sufficient for subsequent statistical analysis and model validation.

4.3 Descriptive Statistical Analysis

This study conducted descriptive statistical analysis on 305 valid samples, with results presented in Table 3.

Table 3. Descriptive Statistical Analysis

project	class	number of people	percentage %
sex	man	152	49.8
	woman	153	50.2
record of formal schooling	Junior high school and below	15	4.9
	High School/Diploma	34	11.1
	junior college education	95	31.1
	undergraduate course	151	49.5
	Postgraduate level or above	10	3.3
	Under 18 years old	20	6.6
age	18-25	46	15.1
	26-30	91	29.8
	31-40	77	25.2
	41-50	37	12.1
	51-60	25	8.2

occupation	Over 60 years old	9	3
	Government agencies (civil servants)	27	8.9
	Public institutions (teachers, doctors, etc.)	53	17.4
	Corporate Employees	122	40
	self-employed businessman	73	23.9
	student	20	6.6
	other	10	3.3
income	Below 3000	28	9.2
	3000-6000	67	22
	6000-9000	135	44.3
	Over 9000	75	24.6

The sample structure shows a balanced gender distribution with 152 males (49.8%) and 153 females (50.2%). Educational attainment reveals that bachelor's degree holders constitute the largest proportion at 49.5%. Age distribution indicates respondents are predominantly aged 26-40, reflecting their as a demographic with established purchasing power and consumption experience. Occupationally, corporate employees account for the highest share at 40.0%, while income distribution shows monthly earnings between 6,000-9,000 yuan as the most common range, representing 44.3% of respondents.

Overall, the sample exhibits favorable distribution characteristics in terms of gender, age, occupation, and income, effectively reflecting the demographic profile of green food consumers.

4.4 Reliability and Validity Analysis

4.4.1 Reliability Analysis

Reliability is primarily used to assess the consistency and stability of measurement results in scales. This study employed Cronbach's α coefficient and corrected item-total correlation (CITC) to evaluate the internal consistency of variable scales. Generally, when the CITC value exceeds 0.5, it indicates strong inter-item correlation; whereas a Cronbach's α coefficient greater than 0.7 signifies good reliability of the scale.

The reliability analysis results of this study are presented in Table 4.

Table 4. Reliability Analysis Results

variable	Number of items	Total correlation of calibration terms (CITC)	Deleted items Cronbach's α coefficient	Cronbach's α coefficient	Overall scale Cronbach's α coefficient
Brand name	A1	0.72	0.819	0.862	0.930
	A2	0.698	0.828		
	A3	0.704	0.826		
	A4	0.713	0.822		
Authentication identifier	B1	0.724	0.818	0.862	
	B2	0.695	0.83		
	B3	0.715	0.822		
	B4	0.702	0.827		
price	C1	0.651	0.799	0.829	
	C2	0.717	0.735		
	C3	0.696	0.756		
Quality Comment	D1	0.738	0.794	0.858	
	D2	0.726	0.805		
	D3	0.729	0.803		
perceived value	M1	0.723	0.827	0.867	

	M2	0.719	0.829	
	M3	0.707	0.834	
	M4	0.719	0.829	
	F1	0.716	0.774	
China consumer attitudes	F2	0.717	0.774	0.843
	F3	0.692	0.797	
	G1	0.715	0.776	
willingness to buy	G2	0.697	0.794	0.844
	G3	0.715	0.777	

The results demonstrated that the Cronbach's α coefficients for brand name, certification logo, price, quality reviews, perceived value, consumer attitudes, and purchase intention were 0.862, 0.862, 0.829, 0.858, 0.867, 0.843, and 0.844, respectively, with the overall scale's Cronbach's α coefficient reaching 0.930. All α coefficients exceeded 0.8, indicating strong internal consistency across the measurement scale.

Further analysis revealed that all item-specific Cronbach's α coefficients (CITC) exceeded 0.5, indicating strong correlations between items and their respective variables. Moreover, removing any single item did not result in significant improvements in Cronbach's α coefficients, suggesting that the current item structure is well-designed and requires no further modifications.

In conclusion, the scale developed in this study demonstrates favorable reliability levels, enabling subsequent analytical applications.

4.4.2 Validity Analysis

(1) Content validity

Content validity is primarily used to assess whether questionnaire items can effectively reflect the connotation of research variables. All measurement items in this study were derived from established scales and underwent moderate revisions based on the context of green food research, thereby demonstrating good content validity.

(2) Structural validity

To further validate the structural validity of the scale, this study employed the KMO test and Bartlett's test for sphericity to assess the suitability of data for factor analysis. It is generally accepted that when the KMO value exceeds 0.7 and the Bartlett's sphericity test yields significant results, the sample data are deemed appropriate for factor analysis.

The test results are shown in Table 5.

Table 5. KMO and Bartlett sphericity test

KMO price	0.917	
	Approximate chi-square	3958.310
Bartlett sphericity test	free degree	276
	conspicuousness	0.000

The results showed a KMO value of 0.917, an approximate chi-square value of 3958.310 from the Bartlett's sphericity test, 276 degrees of freedom, and a significance level of 0.000, indicating strong correlations among variables and suitability for factor analysis.

Subsequently, this study employed principal component analysis to conduct exploratory factor analysis on all items. The total variance explanation results are presented in Table 6.

Table 6. Total Variance Explanation

ingredient	Initial eigenvalue			extract load square sum			sum of squares of rotational loads		
	amount	variance	accumulat	amount to	variance	accumulate %	amount to	variance	accumulate %
	to	percentage	e %		percentage			percentage	

1	9.21	38.374	38.374	9.21	38.374	38.374	2.919	12.164	12.164
2	1.76	7.332	45.706	1.76	7.332	45.706	2.916	12.15	24.314
3	1.674	6.977	52.683	1.674	6.977	52.683	2.883	12.012	36.326
4	1.392	5.801	58.483	1.392	5.801	58.483	2.323	9.679	46.005
5	1.349	5.621	64.104	1.349	5.621	64.104	2.314	9.64	55.645
6	1.25	5.208	69.312	1.25	5.208	69.312	2.246	9.36	65.006
7	1.177	4.905	74.217	1.177	4.905	74.217	2.211	9.211	74.217
8	0.577	2.405	76.622						
9	0.499	2.081	78.703						
10	0.49	2.041	80.744						
11	0.437	1.821	82.565						
12	0.414	1.727	84.292						
13	0.4	1.667	85.959						
14	0.386	1.608	87.567						
15	0.356	1.484	89.051						
16	0.353	1.469	90.52						
17	0.337	1.402	91.922						
18	0.334	1.391	93.313						
19	0.294	1.225	94.538						
20	0.285	1.187	95.725						
21	0.277	1.155	96.88						
22	0.268	1.117	97.997						
23	0.25	1.044	99.04						
24	0.23	0.96	100						

The results demonstrated the extraction of 7 common factors, with a cumulative variance explanation rate of 74.217%, which is consistent with the original questionnaire structure, indicating that the scale exhibits good structural validity.

The factor loading results after rotation are shown in Table 7.

Table 7. Rotation Factor Load Matrix

Item	ingredient						
	1	2	3	4	5	6	7
A1	0.781	0.1	0.214	0.091	0.132	0.065	0.163
A2	0.774	0.12	0.165	0.099	0.11	0.149	0.11
A3	0.788	0.203	0.146	0.094	0.036	0.079	0.124
A4	0.793	0.14	0.05	0.118	0.193	0.127	0.103
B1	0.155	0.781	0.137	0.144	0.137	0.129	0.12
B2	0.126	0.771	0.144	0.089	0.171	0.088	0.132
B3	0.158	0.766	0.168	0.062	0.112	0.111	0.223
B4	0.131	0.795	0.093	0.175	0.106	0.119	0.062
C1	0.213	0.101	0.135	0.127	0.2	0.774	0.058
C2	0.083	0.175	0.198	0.161	0.103	0.798	0.167
C3	0.108	0.146	0.22	0.164	0.076	0.78	0.176
D1	0.136	0.211	0.162	0.807	0.134	0.099	0.149
D2	0.083	0.126	0.147	0.814	0.153	0.195	0.131
D3	0.174	0.118	0.221	0.781	0.112	0.164	0.175

M1	0.137	0.174	0.738	0.163	0.171	0.235	0.094
M2	0.167	0.112	0.815	0.127	0.045	0.108	0.133
M3	0.117	0.17	0.752	0.198	0.146	0.15	0.137
M4	0.196	0.128	0.745	0.098	0.19	0.142	0.197
F1	0.191	0.173	0.11	0.137	0.794	0.122	0.162
F2	0.133	0.136	0.134	0.128	0.803	0.164	0.168
F3	0.118	0.186	0.218	0.125	0.791	0.087	0.082
G1	0.191	0.135	0.118	0.212	0.14	0.156	0.791
G2	0.134	0.181	0.214	0.099	0.206	0.195	0.747
G3	0.192	0.222	0.207	0.172	0.107	0.079	0.775

The table demonstrates that all items exhibit high loading on their respective factors while showing low cross-loadings across other factors, indicating a well-defined structural classification of variables and establishing the scale's robust foundation for both convergent and discriminant properties.

(3) Converge validity

This study further utilized AMOS 23.0 to examine the convergent validity of the scale, primarily assessing indicators such as standardized factor loadings, composite reliability (CR), and average variance extraction (AVE).

The analysis results are shown in Table 8.

Table 8. Results of converge validity analysis

	way		Standardized factor loadings	CR	AVE
A1	<---	Brand name	0.798		
A2	<---	Brand name	0.770		
A3	<---	Brand name	0.772	0.862	0.610
A4	<---	Brand name	0.784		
B1	<---	Authentication identifier	0.799		
B2	<---	Authentication identifier	0.765		
B3	<---	Authentication identifier	0.791	0.862	0.609
B4	<---	Authentication identifier	0.767		
C1	<---	price	0.733		
C2	<---	price	0.825	0.831	0.621
C3	<---	price	0.804		
D1	<---	Quality Comment	0.823		
D2	<---	Quality Comment	0.804	0.858	0.668
D3	<---	Quality Comment	0.825		
M1	<---	perceived value	0.804		
M2	<---	perceived value	0.770		
M3	<---	perceived value	0.776	0.866	0.619
M4	<---	perceived value	0.796		
F1	<---	China consumer attitudes	0.817		
F2	<---	China consumer attitudes	0.810	0.843	0.642
F3	<---	China consumer attitudes	0.776		
G1	<---	willingness to buy	0.803		
G2	<---	willingness to buy	0.792	0.844	0.643
G3	<---	willingness to buy	0.810		

The results demonstrated that the standardized factor loadings of all latent variables exceeded 0.6, with composite

reliability (CR) consistently above 0.7 and all values surpassing 0.8, while the average variance extracted (AVE) remained consistently greater than 0.5. These findings indicate that each latent variable exhibits robust convergent validity, effectively reflecting their respective underlying constructs.

(4) Discrimination validity

Discriminant validity is used to assess the degree of differences between distinct latent variables. In this study, discriminant validity was examined by comparing the root mean square of AVE (Root Mean Square of Average Variance) with the inter-variable correlation coefficient, as shown in Table 9.

Table 9. Discriminant validity analysis results

	willingness to buy	China consumer attitudes	perceived value	Quality Comment	price	Authentication identifier	Brand name
willingness to buy	0.802						
consumer attitude	0.54**	0.801					
perceived value	0.579**	0.524**	0.787				
Quality Comment	0.563**	0.489**	0.556**	0.817			
price	0.539**	0.48**	0.589**	0.541**	0.788		
Authentication identifier	0.555**	0.513**	0.508**	0.487**	0.479**	0.78	
Brand name	0.531**	0.479**	0.515**	0.441**	0.444**	0.486**	0.781

The results indicate that the square root of AVE on the diagonal of each variable is higher than its correlation coefficient with other variables, suggesting good discriminant validity among the latent variables.

(5) Correlation analysis

To further examine the relationships among variables, this study employed SPSS 26.0 to conduct Pearson correlation analyses on brand names, certification marks, price, quality reviews, perceived value, consumer attitudes, and purchase intention. The results are presented in Table 10.

Table 10. Correlation Coefficients of Variables

	Brand name	Authentication identifier	price	Quality Comment	perceived value	China consumer attitudes	willingness to buy
Brand name	1						
Authentication identifier	0.419**	1					
price	0.386**	0.403**	1				
Quality Comment	0.378**	0.418**	0.458**	1			
perceived value	0.443**	0.437**	0.498**	0.479**	1		
China consumer attitudes	0.406**	0.439**	0.409**	0.417**	0.448**	1	
willingness to buy	0.452**	0.471**	0.450**	0.478**	0.495**	0.455**	1

* $p < 0.05$ ** $p < 0.01$

The results demonstrate that brand names exhibit significant positive correlations with certification labels, pricing, quality reviews, perceived value, consumer attitudes, and purchase intent. Certification labels likewise show significant positive correlations with pricing, quality reviews, perceived value, consumer attitudes, and purchase intent. Other variables also demonstrate significant positive correlations. These findings indicate strong interrelationships among the seven latent variables examined in this study, providing a solid foundation for subsequent hypothesis testing.

4.5 hypothesis test

4.5.1 Impact of External Clues on Purchase Intention

To examine the impact of brand name, certification logo, price, and quality reviews on purchase intention, this study

conducted a multiple linear regression analysis with purchase intention as the dependent variable and four types of external cues as independent variables. The results are presented in Table 11.

Table 11. Regression Results of External Clues on Purchase Intention

	Unstandardized coefficient		Standardization coefficient	t	conspicuousness	collinearity statistics	
	B	Standard error	Beta			tolerance	VIF
(constant)	0.376	0.223		1.685	0.093		
Brand name	0.225	0.058	0.205	3.906	0.000	0.745	1.342
Authentication identifier	0.242	0.059	0.218	4.077	0.000	0.716	1.396
price	0.191	0.057	0.179	3.328	0.001	0.708	1.412
Quality Comment	0.23	0.055	0.228	4.216	0.000	0.704	1.421
adjusted R-squared				0.377			
F				46.926			
P				0.000			

Dependent variable: Purchase intention

The results indicate that the adjusted R^2 of the model is 0.377, demonstrating that brand name, certification logo, price, and quality reviews account for 37.7% of the variance in purchase intention. The F-value is 46.926 with a p-value of 0.000, confirming the overall model's statistical significance. Additionally, all variables exhibit VIF values below 10 and tolerance coefficients above 0.1, indicating no significant multicollinearity issues in the model.

From the regression coefficients of various variables, the standardization coefficient of brand name is 0.205, certification mark is 0.218, price is 0.179, and quality review is 0.228, with all significance levels being less than 0.05. This indicates that all four types of external cues have a significant positive impact on the purchase intention of green food among China consumers.

Therefore, we assume that H1, H2, H3, and H4 are all supported.

4.5.2 Impact of perceived value on purchase intention

To examine the direct impact of perceived value on purchase intention, this study conducted a regression analysis with purchase intention as the dependent variable and perceived value as the independent variable, as shown in Table 12.

Table 12. Regression Results of Perceived Value on Purchase Intention

	Unstandardized coefficient		Standardization coefficient	t	conspicuousness	collinearity statistics	
	B	Standard error	Beta			tolerance	VIF
(constant)	1.599	0.184		8.705	0		
perceived value	0.52	0.052	0.495	9.917	0	1	1
adjusted R-squared				0.243			
F				98.345			
P				0.000			

Dependent variable: Consumer purchase intention

The results showed that the adjusted R^2 of the model was 0.243, with an F-value of 98.345 and a p-value of 0.000, indicating overall statistical significance of the model. The standardized regression coefficient for perceived value was 0.495, with a significance level below 0.05, demonstrating a significant positive impact of perceived value on purchase intention.

In other words, when consumers develop a higher level of value perception towards green foods, their purchase intention will also significantly increase. Therefore, Hypothesis H5 is supported.

4.6 Mediation Test of Perceived Value

To examine the mediating role of perceived value between brand names, certification marks, price and quality reviews and purchase intention, this study conducts stratified regression analyses on four distinct pathways.

4.6.1 Brand Name, Perceived Value, and Purchase Intention

The regression results between brand name, perceived value, and purchase intention are presented in Table 13.

Table 13. Regression Results Between Brand Name, Perceived Value, and Purchase Intention

	willingness to buy		perceived value		willingness to buy	
	Standardized coefficient Beta	t	Standardized coefficient Beta	t	Standardized coefficient Beta	t
Brand name	0.452	8.811***	0.443	8.608***	0.289	5.427***
perceived value					0.367	6.892***
R Fang	0.204		0.197		0.312	
adjusted R-squared	0.201		0.194		0.308	
F	77.627		74.106		68.515	

The results indicate that brand name accounts for 20.4% of purchase intention explanatory power, which increases to 31.2% when perceived value is included. Meanwhile, the regression coefficient of brand name on purchase intention decreases from 0.452 to 0.289 while remaining statistically significant, demonstrating that perceived value partially mediates the relationship between brand name and purchase intention.

Therefore, if H5a is supported.

4.6.2 Certification Marking, Perceived Value, and Purchase Intention

The regression results between certification labels, perceived value, and purchase intention are presented in Table 14.

Table 14. Regression Results Between Certification Label, Perceived Value, and Purchase Intention

	willingness to buy		perceived value		willingness to buy	
	Standardized coefficient Beta	t	Standardized coefficient Beta	t	Standardized coefficient Beta	t
Authentication identifier	0.471	9.303***	0.437	8.467***	0.315	5.996***
perceived value					0.357	6.795***
R Fang	0.222		0.191		0.325	
adjusted R-squared	0.220		0.189		0.321	
F	86.552		71.693		72.818	

The results indicate that certification labels account for 22.2% of purchase intention explanatory power, which increases to 32.5% when perceived value is included. Meanwhile, the regression coefficient of certification labels on purchase intention decreases from 0.471 to 0.315 while remaining statistically significant, demonstrating that perceived value partially mediates the relationship between certification labels and purchase intention.

Therefore, if H5b is supported.

4.6.3 Price, Perceived Value, and Purchase Intention

The regression results between price, perceived value, and purchase intention are presented in Table 15.

Table 15. Regression Results Between Price, Perceived Value, and Purchase Intention

	willingness to buy		perceived value		willingness to buy	
	Standardized coefficient Beta	t	Standardized coefficient Beta	t	Standardized coefficient Beta	t

price	0.450	8.782***	0.498	10.006***	0.271	4.882***
perceived value					0.360	6.483***
R Fang	0.203		0.248		0.300	
adjusted R-squared	0.200		0.246		0.296	
F	77.120		100.123		64.794	

The results indicate that price accounts for 20.3% of purchase intention explanatory power, which increases to 30.0% when perceived value is incorporated into the model. Meanwhile, the regression coefficient of price on purchase intention decreases from 0.450 to 0.271 while remaining statistically significant, demonstrating that perceived value partially mediates the relationship between price and purchase intention.

Therefore, if H5c is supported.

4.6.4 Quality Reviews, Perceived Value, and Purchase Intention

The regression results between quality reviews, perceived value, and purchase intention are shown in Table 16.

Table 16. Regression results between quality comments, perceived value, and purchase intention

	willingness to buy		perceived value		willingness to buy	
	Standardized coefficient Beta	t	Standardized coefficient Beta	t	Standardized coefficient Beta	t
Quality Comment	0.478	9.482***	0.479	9.501***	0.313	5.793***
perceived value					0.345	6.385***
R Fang	0.229		0.230		0.321	
adjusted R-squared	0.226		0.227		0.316	
F	89.902		90.263		71.236	

The results indicate that quality reviews account for 22.9% of purchase intention explanatory power, which increases to 32.1% when perceived value is included. Meanwhile, the regression coefficient of quality reviews on purchase intention decreases from 0.478 to 0.313 while remaining statistically significant, demonstrating that perceived value partially mediates the relationship between quality reviews and purchase intention.

Therefore, if H5d is supported.

Overall, perceived value partially mediates the relationship between four types of external cues and purchase intention.

4.7 Adjustment Effect Test

To examine the moderating role of consumer attitudes in the relationship between external cues and perceived value, this study constructs interaction terms between external cues and consumer attitudes and conducts hierarchical regression analysis.

4.7.1 Brand Name, Consumer Attitude, and Perceived Value

The regression results between brand name, consumer attitude, and perceived value are presented in Table 17.

Table 17. Regression Results of Brand Name, Consumer Attitude, and Perceived Value

	perceived value			
	Standardized coefficient Beta	t	Standardized coefficient Beta	t
Brand name	0.443	8.608***	0.28	5.324***
consumer attitude			0.288	5.48***
Brand Name*Consumer Attitude			0.205	4.16***
R Fang	0.197		0.322	
adjusted R-squared	0.194		0.315	
F	74.106		47.567	

The results indicate that the adjusted R^2 of the regression model between brand name and perceived value was 0.194.

After incorporating consumer attitudes and interaction terms, the adjusted R^2 increased to 0.315. The standardized regression coefficient of brand name $\times$ consumer attitude was 0.205, which is statistically significant, demonstrating that consumer attitudes exert a significant positive moderating effect between brand name and perceived value.

Therefore, if H6a is supported.

4.7.2 Certification Marking, Consumer Attitude, and Perceived Value

The regression results between certification labels, consumer attitudes, and perceived value are presented in Table 18.

Table 18. Regression Results of Certification Mark, Consumer Attitude, and Perceived Value

	perceived value			
	Standardized coefficient Beta	t	Standardized coefficient Beta	t
Authentication identifier	0.437	8.467***	0.242	4.572***
consumer attitude			0.295	5.664***
Certification Label*Consumer Attitude			0.275	5.711***
R Fang	0.191		0.344	
adjusted R-squared	0.189		0.337	
F	71.693		52.566	

The results indicate that the adjusted R^2 of the regression model for certification labels and perceived value was 0.189. When consumer attitudes and interaction terms were included, the adjusted R^2 increased to 0.337. The standardized regression coefficient of certification labels $\times$ consumer attitudes was 0.275, which was statistically significant, demonstrating that consumer attitudes exert a significant positive moderating effect between certification labels and perceived value.

Therefore, if H6b is supported.

4.7.3 Price, Consumer Attitude, and Perceived Value

The regression results between price, consumer attitudes, and perceived value are presented in Table 19.

Table 19. Regression Results of Price, Consumer Attitude, and Perceived Value

	perceived value			
	Standardized coefficient Beta	t	Standardized coefficient Beta	t
price	0.498	10.006***	0.363	7.005***
China consumer attitudes			0.274	5.247***
Price*China consumer attitude			0.122	2.528*
R Fang	0.248		0.334	
adjusted R-squared	0.246		0.328	
F	100.123		50.370	

The results indicate that the adjusted R^2 of the regression model between price and perceived value was 0.246. When consumer attitudes and interaction terms were incorporated, the adjusted R^2 increased to 0.328. The standardized regression coefficient of price $\times$ consumer attitudes was 0.122 ($p < 0.05$), demonstrating that consumer attitudes exert a significant positive moderating effect between price and perceived value.

Therefore, if H6c is supported.

4.7.4 Quality Reviews, Consumer Attitudes, and Perceived Value

The regression results between quality reviews, consumer attitudes, and perceived value are shown in Table 20.

Table 20. Regression Results of Quality Reviews, Consumer Attitudes, and Perceived Value

	perceived value			
	Standardized coefficient Beta	t	Standardized coefficient Beta	t
Quality Comment	0.479	9.501***	0.335	6.535***

consumer attitude		0.265	5.11***
Quality Reviews*Consumer Attitudes		0.217	4.565***
R Fang	0.230	0.349	
adjusted R-squared	0.227	0.343	
F	90.263	53.874	

The results indicate that the adjusted R^2 of the regression model for quality reviews and perceived value was 0.227. When consumer attitudes and interaction terms were included, the adjusted R^2 increased to 0.343. The standardized regression coefficient of quality reviews $\times$ consumer attitudes was 0.217, which was statistically significant, demonstrating that consumer attitudes exert a significant positive moderating effect between quality reviews and perceived value.

Therefore, if H6d is supported.

Overall, consumer attitudes exhibit significant positive moderating effects across all four categories of external cues and perceived value.

4.8 Summary of Research Hypothesis Testing Results

Based on the aforementioned empirical analysis results, all research hypotheses proposed in this study have been supported. The specific findings are presented in Table 21.

Table 21. Summary of Study Hypothesis Testing Results

research hypothesis	Approve or reject
H1: Brand name has a positive impact on consumers' purchase intention for green food.	pass through
H2: The certification label positively influences consumers' purchase intention for green food.	pass through
H3: Price exerts a positive influence on consumers' purchase intention for green food.	pass through
H4: Quality reviews positively influence the purchase intention for green foods.	pass through
H5: Perceived value positively influences the purchase intention for green foods.	pass through
H5a: Perceived value mediates the relationship between brand name and purchase intention.	pass through
H5b: Perceived value mediates the relationship between certification labels and purchase intention.	pass through
H5c: Perceived value mediates the relationship between price and purchase intention.	pass through
H5d: Perceived value mediates the relationship between quality reviews and purchase intention.	pass through
H6a: China consumers' attitudes play a moderating role between brand effect and perceived value.	pass through
H6b: China consumers' attitudes play a moderating role between certification labels and perceived value.	pass through
H6c: China consumers' attitudes play a moderating role between price and perceived value.	pass through
H6d: China consumers' attitudes play a moderating role between quality reviews and perceived value.	pass through

5 Discussion and Conclusion

5.1 Research Conclusions and Discussion

Based on cue utilization theory and combined with the stimulus-organism-response (S-O-R) theoretical framework, this study systematically examines the influence mechanism of external cues on Chinese consumers' green food purchase intention. In the research design, brand name, certification labels, price, and quality reviews are defined as external stimulus variables, perceived value as a mediating variable, and consumer attitudes as a moderating variable, constructing a theoretical model of green food purchase intention. Based on the empirical analysis results of 305 valid questionnaires, the following conclusions are primarily obtained.

The research results indicate that external cues significantly positively influence Chinese consumers' purchase intention for green food. Among these, the impact of brand name, certification labels, price, and quality reviews all reached significant levels, with quality reviews having the strongest effect, followed by certification labels, and price being relatively weaker. This result suggests that in the context of green food consumption, Chinese consumers particularly value information cues related

to authentic product usage experiences, especially in online consumption environments where product reviews have become a crucial basis for consumers to assess product quality, safety, and credibility. Compared to traditional price judgments, Chinese consumers tend to rely more on review information to reduce purchase uncertainty and make purchasing decisions accordingly. This also reflects, to some extent, that with changes in consumption environments and information dissemination methods, consumers' value judgments about green food increasingly depend on accessible, verifiable, and socially interactive external information.

The study further revealed that perceived value partially mediates the relationship between external cues and green food purchase intention. This indicates that external cues do not directly and completely translate into purchase intent, but first influence consumers' overall value judgment of green foods, thereby driving the formation of purchase intention. In other words, brand names, certification labels, pricing, and quality reviews affect purchase intention because these cues enhance consumers' value perception of green foods in terms of safety, nutritional value, health benefits, and overall quality. Simultaneously, perceived value is not the sole mediating pathway—the impact of external cues on purchase intention may also be influenced by other psychological and cognitive variables such as green trust, green knowledge, or product knowledge. Therefore, the findings not only validate the critical role of perceived value in green food consumption decisions, but also demonstrate that the formation of green food purchase intention constitutes a complex process involving multiple interacting factors.

Furthermore, consumer attitudes play a significant positive moderating role between external cues and perceived value. When consumers hold more positive attitudes toward green foods, external cues exert a stronger positive influence on perceived value. This indicates that consumers do not passively accept external information but actively filter, interpret, and process external cues based on their existing attitudes. For consumers with more positive attitudes, information such as brand reputation, certifications, pricing, and reviews is more readily perceived as positive value attributes, thereby enhancing their overall perception of green food value. These findings demonstrate that consumer attitudes not only serve as a critical precursor to green consumption behaviors but also play a pivotal role in transforming external information into value judgments. This discovery deepens our understanding of green food consumption decision-making mechanisms and highlights that the formation of green consumption behaviors relies not only on objective informational cues but is also profoundly influenced by consumers' subjective psychological tendencies.

Overall, this study employs clue utilization theory and the S-O-R theory to validate the fundamental pathway of "external cues—perceived value—purchase intention" in green food consumption, while further identifying the reinforcing role of consumer attitudes. The findings demonstrate that green food purchase intention is not solely determined by product attributes or price levels, but rather emerges through the combined effects of external information stimuli, internal value perception, and individual attitudinal tendencies. This research not only provides new empirical support for green food consumption behavior studies, but also establishes a theoretical foundation for future investigations into green consumption decision-making mechanisms.

5.2 Management Insights

The empirical findings of this study provide valuable insights for market practices in green food enterprises. Firstly, companies should prioritize the role of external cues in consumer purchasing decisions, particularly through systematic strategies in brand building, certification communication, pricing management, and social media engagement. Research indicates that both brand names and certification labels significantly enhance purchase intent. This suggests that businesses must not only establish clear, credible, and recognizable brand identities but also optimize green certification presentation methods and communication effectiveness to reduce information uncertainty during purchasing decisions. For trust-driven products like green foods, transparent, authoritative, and verifiable information cues often prove more effective in building consumer trust than simplistic promotional slogans.

Secondly, businesses should focus on enhancing consumers' perceived value of green foods. Research indicates that perceived value serves as a critical intermediary between external cues and purchase intent. Therefore, companies must go beyond mere information presentation in marketing strategies and actively guide consumers to recognize the comprehensive

value of green foods in terms of safety, nutritional benefits, health benefits, and quality. For instance, enterprises can employ persuasive product descriptions, visual traceability information, transparent ingredient displays, and quality assurance commitments to help consumers develop clearer and more positive value perceptions. Only when consumers genuinely feel that green foods are "worth purchasing" can the information conveyed through external cues be effectively translated into actual purchase intent.

Thirdly, enterprises should prioritize value alignment and market segmentation in pricing strategies. Research findings indicate that while price exerts a significant positive impact on purchase intent, its effect is relatively weaker compared to quality evaluations and certification labels. This demonstrates that in green food consumption, price is not the sole determining factor—consumers place greater emphasis on whether product quality and value correspond to pricing levels. Therefore, green food enterprises should emphasize the cognitive logic of "price matching value" when formulating pricing strategies. For premium green food brands, moderate price premiums can effectively communicate signals of high quality and safety. For mass-market green food products, however, avoiding high prices coupled with low quality is crucial to prevent negative consumer perceptions, thereby safeguarding brand trust and maintaining long-term market competitiveness.

Finally, businesses should prioritize comment management and digital reputation building. Our study reveals that quality reviews have the most significant impact on purchase intent, indicating that in the digital consumption landscape, review information has become a critical factor in consumers' decision-making process regarding green food products. Companies should encourage authentic user participation in high-quality feedback, optimize review display mechanisms, and proactively address consumer inquiries and concerns in comment sections. Compared to one-way advertising messages, reviews based on genuine experiences are more effective in enhancing potential customers' perceived value and purchase confidence. For green food enterprises, establishing a credible, transparent, and interactive review ecosystem will serve as a key strategy to improve market conversion rates.

5.3 Research Limitations and Future Prospects

Although this study has conducted a relatively systematic analysis of the formation mechanism of green food purchase intention and achieved certain research findings, several limitations remain, requiring further refinement in future research.

Firstly, this study primarily examines external cues through four dimensions: brand names, certification labels, pricing, and quality reviews. While these four categories are highly representative in green food consumption, external information cues for green products extend beyond them. Advertising communication, packaging design, origin information, and sales channel image may also significantly influence consumer judgment. Future research should therefore expand the scope of external cue dimensions to comprehensively evaluate the relative influence and interaction boundaries of different cues in green food consumption decision-making processes.

Secondly, this study primarily focuses on the impact of external cues on purchase intention, while internal cues receive relatively insufficient attention. In reality, consumers purchasing green foods are influenced not only by external factors such as brand reputation, pricing, and product reviews, but also by internal cues including taste, color, and aroma. Future research could integrate both internal and external cues into a unified analytical framework to further compare their respective roles in green food consumption decision-making. By exploring potential interactions between different cue types, we can gain deeper insights into the underlying mechanisms shaping green food consumption behaviors.

Finally, the data in this study were primarily collected through online questionnaires. While this method offers high convenience and efficient sample acquisition, it remains challenging to fully control respondents' answering environments, levels of attention engagement, and contextual interference factors. Additionally, cross-sectional questionnaire data inherently possess limitations in causal inference. Future research could incorporate experimental methods, longitudinal tracking surveys, or multi-source data collection approaches to enhance the robustness and explanatory power of conclusions. Comparative studies involving different regions, consumer demographics, or green food categories could also be conducted to strengthen the generalizability and external validity of findings.

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